

1. Transformer works on the principle of:

- A) Self induction
- B) Mutual induction
- C) Capacitance
- D) Resistance

Answer: B

2. Transformer changes:

- A) Frequency
- B) Voltage level
- C) Power
- D) Current only

Answer: B

3. Efficiency of ideal transformer is:

- A) 50%
- B) 75%
- C) 100%
- D) 0%

Answer: C

4. Core of transformer is made of:

- A) Copper
- B) Aluminium
- C) Laminated iron
- D) Steel

Answer: C

5. DC machine works on:

- A) Faraday's law
- B) Ohm's law
- C) Kirchhoff law
- D) Coulomb law

Answer: A

6. Commutator is used in:

- A) Transformer
- B) AC motor
- C) DC machine
- D) Alternator

Answer: C

7. Function of commutator:

- A) Increase voltage
- B) Convert AC to DC
- C) Reduce current
- D) Store energy

Answer: B

8. Induction motor works on:

- A) Mutual induction
- B) Self induction
- C) Capacitance
- D) Resistance

Answer: A

9. Slip in induction motor is:

- A) Zero

B) Negative
C) Difference between speeds
D) Infinite
Answer: C

10. Rotor speed is always:
A) Greater than synchronous
B) Equal
C) Less
D) Zero
Answer: C

11. Synchronous motor speed is:
A) Variable
B) Constant
C) Zero
D) Infinite
Answer: B

12. Alternator converts:
A) Electrical to mechanical
B) Mechanical to electrical
C) Heat to electrical
D) None
Answer: B

13. Frequency of alternator depends on:
A) Voltage
B) Speed
C) Current
D) Resistance
Answer: B

14. EMF is induced due to:
A) Resistance
B) Magnetic field change
C) Voltage
D) Power
Answer: B

15. Fleming's right hand rule is used for:
A) Motor
B) Generator
C) Transformer
D) Battery
Answer: B

16. Fleming's left hand rule is used for:
A) Generator
B) Motor
C) Transformer
D) Alternator
Answer: B

17. Losses in transformer include:
A) Copper loss
B) Iron loss
C) Both

D) None

Answer: C

18. Core loss is also called:

A) Copper loss

B) Iron loss

C) Mechanical loss

D) Electrical loss

Answer: B

19. Copper loss occurs in:

A) Core

B) Windings

C) Frame

D) Rotor

Answer: B

20. Step-up transformer:

A) Decreases voltage

B) Increases voltage

C) Same voltage

D) Zero voltage

Answer: B

21. Step-down transformer:

A) Increase voltage

B) Decrease voltage

C) Same voltage

D) Zero

Answer: B

22. Armature winding is in:

A) Rotor

B) Stator

C) Both

D) None

Answer: A

23. Field winding produces:

A) Voltage

B) Magnetic field

C) Current

D) Power

Answer: B

24. Brushes are used to:

A) Supply current

B) Increase voltage

C) Reduce current

D) Store energy

Answer: A

25. DC generator output is:

A) AC

B) DC

C) Both

D) None

Answer: B

26. Slip rings are used in:

- A) DC machines
- B) AC machines
- C) Both
- D) None

Answer: B

27. Induction motor rotor type:

- A) Squirrel cage
- B) Slip ring
- C) Both
- D) None

Answer: C

28. Synchronous speed formula includes:

- A) Voltage
- B) Frequency
- C) Resistance
- D) Power

Answer: B

29. Transformer works on:

- A) DC supply
- B) AC supply
- C) Both
- D) None

Answer: B

30. Hysteresis loss depends on:

- A) Voltage
- B) Frequency
- C) Resistance
- D) Current

Answer: B

31. Eddy current loss is reduced by:

- A) Thick core
- B) Laminations
- C) Copper
- D) Aluminium

Answer: B

32. Power factor of motor:

- A) Unity
- B) Lagging
- C) Leading
- D) Zero

Answer: B

33. Starting torque of induction motor is:

- A) High
- B) Low
- C) Zero
- D) Infinite

Answer: B

34. Rotor in alternator is:

- A) Stationary
- B) Rotating
- C) Fixed
- D) None

Answer: B

35. Stator in alternator is:

- A) Rotating
- B) Stationary
- C) Moving
- D) None

Answer: B

36. Efficiency means:

- A) Input/output
- B) Output/input
- C) Power only
- D) Voltage

Answer: B

37. Back EMF occurs in:

- A) Generator
- B) Motor
- C) Transformer
- D) Battery

Answer: B

38. Back EMF opposes:

- A) Voltage
- B) Current
- C) Supply
- D) Resistance

Answer: C

39. Losses in DC machine:

- A) Copper
- B) Mechanical
- C) Both
- D) None

Answer: C

40. Transformer oil is used for:

- A) Cooling
- B) Insulation
- C) Both
- D) None

Answer: C

41. Frequency of supply in India:

- A) 60 Hz
- B) 50 Hz
- C) 25 Hz
- D) 100 Hz

Answer: B

42. AC motor is:

- A) DC machine
- B) Induction motor

C) Battery
D) Transformer
Answer: B

43. Generator converts:
A) Electrical to mechanical
B) Mechanical to electrical
C) Heat
D) None
Answer: B

44. Motor converts:
A) Mechanical to electrical
B) Electrical to mechanical
C) Heat
D) None
Answer: B

45. Slip is zero in:
A) Induction motor
B) Synchronous motor
C) DC motor
D) Transformer
Answer: B

46. Rotor bars are in:
A) Core
B) Frame
C) Squirrel cage
D) Brush
Answer: C

47. Field current controls:
A) Voltage
B) Speed
C) Resistance
D) Power
Answer: A

48. Transformer rating is in:
A) kW
B) kVA
C) Volt
D) Ampere
Answer: B

49. DC supply is used in:
A) Transformer
B) Induction motor
C) Battery
D) None
Answer: C

50. Electrical machine converts energy:
A) One form to another
B) Only electrical
C) Only mechanical
D) None

Answer: A